

Robert Rothschild

rarothisc@gmail.com | 720-785-0488 | Bellingham, WA 98226
robthischild.com | linkedin.com/in/rob-rothschild

SUMMARY

Systems Engineer with 8 years of experience in the space sector. Focused on delivering software for MBSE, robotics, HIL testbeds, satellite imagery, and scientific computing. Comfortable designing architectures, shipping APIs & GUIs, automating workflows, and collaborating across software, hardware, and I&T teams.

Languages: Python, MATLAB, JS/TS, Rust, C++, SQL, HTML5, CSS

Libraries: NumPy, Pandas, SciPy, D3.js, Plotly

DevOps: Git, Docker, Kubernetes, Google Cloud, CI/CD

Domains: MBSE tooling, scientific computing, statistical requirement verification, data visualization

EXPERIENCE

Systems Engineer (Software Focus) - Heliospace

Mar 2017–Jul 2022; Jan 2023–Jul 2025

- Defined, managed, and verified hundreds of requirements related to most fields of Aerospace projects including mechanical, electrical, software, and thermal.
- Involved in the design of mechanisms and structures including deployable large scale space telescopes, small scale cube sats, and various restraint & release mechanisms.
- Reviewed and performed stress/margin checks on hardware across programs (JWST, RST, HERMES, REASON). Processed FEM outputs to screen stress hot-spots and fastener loads; correlated FEA predictions to test data (random/sine vibe, sine burst) and inserted results into MathCAD-ready reports.
- Oversaw I&T activities including mechanism deployments/stowing, TVAC, vibrations, and load tests at unit and systems levels.
- Cross-checked materials/allowables, torque specs, and thermal properties, edited and checked Electrical ICDs, and contributed to verification matrix closeout and mission assurance activities.
- Built MBSE data pipelines enabling analysis workflows for numerous modeling software including stress, micro-meteoroid hyper-velocity impacts, and space based mission planning tools
- Built Python/MATLAB GUIs that automated requirement verification, stress/margins reports, and model-to-test comparisons, reducing analysis turnaround time for engineering teams.
- Created tooling to compare FEA predictions with test data (random/sine vibe, sine burst) and to screen critical hardware components and locations, then packaged results into engineer-ready reports.
- Independently analyzed diverse system datasets (model outputs, test logs and results) to distill key findings and present clear summaries to cross-functional stakeholders and program leadership.
- Ensured continuous, secure connectivity across mixed remote and local compute nodes (VPN, routing, access controls), enabling sustained parallel execution of hundreds of engineering simulations.
- Supported internal planning and risk identification by validating models against test data, reviewing stress reports and material properties, and contributing to mission assurance.
- Authored and delivered technical presentations, design reviews, and documentation to cross-functional stakeholders. Translated modeling insights into actionable decisions for design and test.

Projects and Missions Impact

- Contributions supported: James Webb Space Telescope (JWST), Mars Sample Return (MSR), Roman Space Telescope (RST), Heliophysics Environmental and Radiation Measurement Experiment Suite (HERMES), Europa Clipper – REASON

Research Engineer - Colorado Space Grant Consortium / Fort Lewis College

2015 – 2018

- Designed, fabricated, and programmed an autonomous desert rover and multiple high-altitude balloon payloads (electric field, radiation, temperature). Operated robotic systems (Baxter) and an autonomous survey boat then processed data used in peer-reviewed publication.

EDUCATION

B.S in Engineering - Fort Lewis College, Durango CO

2013 - 2017

PUBLICATIONS

- Yoo, S.-H., Stuntz, A., Zhang, Y., **Rothschild, R.**, Hollinger, G. A., & Smith, R. N. (2016). Experimental Analysis of Receding Horizon Planning Algorithms for Marine Monitoring. Springer Tracts in Advanced Robotics, 31–44. https://doi.org/10.1007/978-3-319-27702-8_3

AWARDS AND HONORS

- Best Obstacle Avoidance - Colorado Space Grant Consortium's Robotics Challenge
- Finalist - Hawk Tank Business Plan Competition - Fort Lewis College
- Sigma-Pi-Sigma Physics Honor Society Member
- Presenter at the 2015 AIAA Annual Rocky Mountain Technical Symposium
- Technology Award - Chaparral High School - Parker, CO